Does Multi-Agent Tool Calling Need Knowledge-Infused Learning?

A policy graph used at three KiL depths vs. a runtime validator on τ-bench and τ²-bench

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Code & data

1 The problem

An agent gets the airline policy in its prompt and calls tools to book, change, or cancel flights.

pass^k: a task counts only if the agent solves it in all k tries.

0.525 → **0.360**

pass¹ (baseline) pass⁸ (baseline)

Task 14, same task, two tries:

step	PASS (trial 2)	FAIL (trial 1)
Budget	user: "I have a budget of $600"	user: "as long as it doesn't exceed $600"
Next	agent: search_direct_flight(…)	agent: "I don't have a tool to get the price difference"
End	agent: update_reservation_baggages (…)	user: "please revert the changes"

2 Questions

RQ1: Does a policy graph (structured knowledge) make the agent more stable than prose?

RQ2: Does the depth of use matter: shallow, semi-deep, deep?

H1: KiL gives a flatter pass^k curve than a validator that holds the same rules as code.

3 Setup

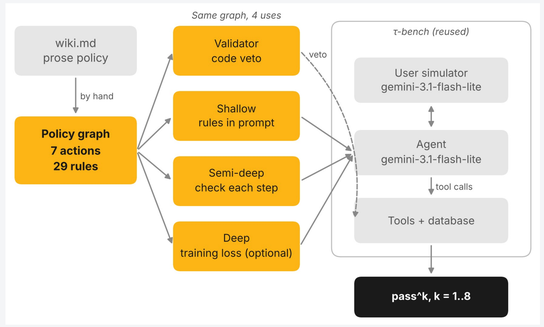
Data: τ-bench airline, 50 tasks (42 clean); τ²-bench telecom, 40 tasks.

Models: gemini-3.1-flash-lite (agent + user); Qwen3-8B for deep KiL.

Runs: 8 trials per task; 12 arms; 4,560 dialogues.

Stats: paired bootstrap over tasks, 2,000 samples; * = 95% CI excludes 0.

4 One graph, four uses



Policy graph: wiki.md by hand → 7 actions, 29 rules (telecom: 57 entries).

Validator (control): 24 data rules as code; blocks an illegal call.

Shallow: graph text added to the prompt.

Semi-deep: before each write call, show that action's rules; agent checks each.

Deep: LoRA on Qwen3-8B + penalty −log(1−p) on rule-breaking calls.

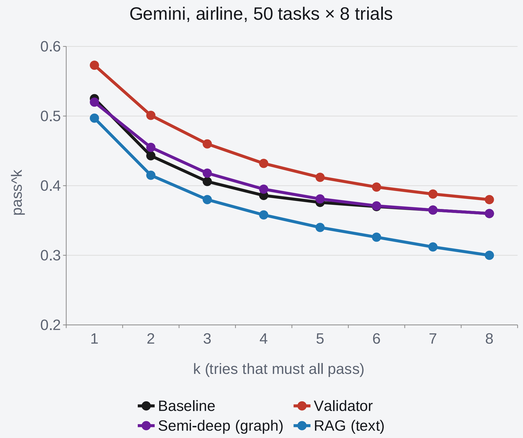
RAG (ablation): semi-deep, but rules come from vector search over the policy text.

5 Airline: all arms

Model	Arm	pass¹	pass⁸	Illegal calls run
Gemini	Baseline	0.525	0.360	70
	Validator	**0.573**	**0.380**	0 (69 blocked)
	Shallow	0.520	0.340	31
	Semi-deep	0.520	0.360	2
	RAG	0.497	0.300	28
Qwen3-8B	Base	0.147	0.020	349
	SFT	0.253	0.040	182
	Deep KiL	0.340	0.120	76
	Validator	**0.390**	**0.200**	0 (407 blocked)

Best arm per model in bold. Validator − baseline, clean 42: pass¹ +0.104*. Deep − SFT: pass¹ +0.104*, pass⁸ +0.095*.

6 Airline: the validator wins

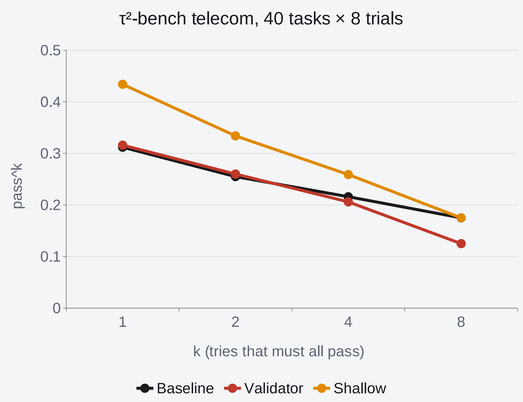


Validator has the highest curve; shallow and semi-deep end at the baseline.

No KiL curve is flatter (drop 1→8: 31–35%).

KiL − validator, clean 42: shallow −0.092*, semi-deep −0.065*.

7 Telecom: shallow wins once



Shallow − validator: pass¹ +0.119*, pass² +0.074*, pass⁸ n.s.

Gain is on mobile data (0.167 → 0.472): the graph adds explicit fix steps.

The validator sees only 11% of answer-key calls; most fixes are on the user's phone.

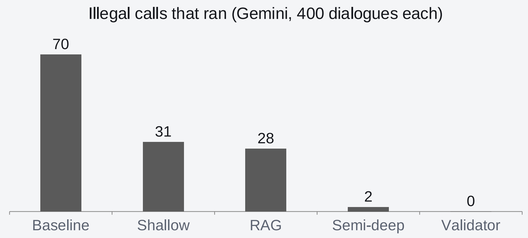
8 Ablation: graph vs text RAG

Same check, same prompt, same amount of text. Only the source of the rules changes.

2 violations semi-deep (graph)

28 violations RAG (text)

71% rules RAG retrieved



pass^k: no significant difference (clean 42, pass¹ −0.036).

24 of 28 violations break C3, the rule RAG missed most for cancellations.

9 Failure analysis

Violations down, failures not: missing actions rise (Gemini 110 → 151); the agent gets too careful.

Score hides violations: 13% of airline passes and 46% of telecom passes still break a rule.

Hand-check of 45 fails: airline agent 20, simulator 5, benchmark 4; telecom all agent.

Benchmark: 8 of 50 airline answer keys break the policy (0 in τ²-bench).

KiL failure types: process, grounding, reasoning; RAG adds retrieval.

10 Takeaways

H1 refuted: rules as code beat every KiL depth on airline.

The graph keeps the agent inside the policy (70 → 2 violations), but does not raise pass^k.

Structure matters for safety: text retrieval misses rules, and a missed rule becomes a broken one.

[1] Yao et al. 2024, τ-bench. [2] Barres et al. 2025, τ²-bench. [3] Gaur & Sheth 2026, Knowledge-Infused Learning. [4] Sheth, Roy & Gaur 2023, Neurosymbolic AI. [5] Roy et al. 2023, ProKnow. [6] Hu et al. 2021, LoRA. · github.com/zhang-systems/policy-graph-agents